INCH-POUND

MIL-E-24416B(SH)
8 February 1991

SUPERSEDING
MIL-E-24416A(SH)
22 June 1979

MILITARY SPECIFICATION

EXTINGUISHING SYSTEM, FIRE, FIXED, AQUEOUS POTASSIUM CARBONATE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes component requirements for chemical fire extinguishing systems of the fixed (stationary) type whereby an aqueous potassium carbonate (K^2CO^3) solution (stowed in cylinders pressurized to 175 pounds per square inch (lb/in^2), gage at 70 degrees Fahrenheit (°F) (21.1 degrees Celsius (°C)) with nitrogen) is expelled automatically or manually through a fixed piping system and suitably designed nozzles into burning deep fat fryers to extinguish the fire by saponification of the surface fat.

1.2 Classification. Extinguishing systems shall be of the following classes and types as specified (see 6.2.1):

Class I – Systems in which magnetic materials are permitted

Class II – Systems in which all materials are non-magnetic

Type A – One 2-1/2-quart capacity cylinder

Type B – One 4-gallon capacity cylinder

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4210

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-C-96	Cans, Metal, 28 Gage and Lighter
PPP-C-569	Container, Plastic, Molded (For Liquids, Pastes and Powders): Overpacked
PPP-F-320	Fiberboard, Corregated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-P-704	Pail Metal: (Shipping, Steel 1 through 12 Gallons)
RR-C-901/1	Cylinder, Compressed Gas; DOT Specification 3AA and Aluminum Applications, for Industrial Services
RR-C-910	Cylinders, Compressed Gas; DOT 4BA, DOT 4BW, and DOT 4E

MILITARY

MIL-P-116	Preservation, Methods of
MIL-S-901	Shock tests, HI (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for
MIL-P-15024	Plates, Tags and Bands for Identification of Equipment
MIL-P-15024/5	Plates, Identification
MIL-S-16032	Switches and Detectors, Shipboard Alarm Systems
MIL-C-16310	Cylinder, Compressed Gas (Compressed Air and Carbon Dioxide, Nonshatterable and Nonmagnetic)

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MIL-I-17214	Indicator, Permeability; Low Mu (Go – No Go)
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Provisional Items (Repair Parts); Packaging of
MIL-L-19140	Lumber and Plywood, Fire Retardant Treated

STANDARDS

DOD-STD-2142	Magnetic Silencing Characteristics, Measurement of; (METRIC)
DOD-STD-2143	Magnetic Silencing Requirements for the Construction of Nonmagnetic Ships and Craft; (METRIC)

FEDERAL

FED-STD-595	Colors Used in Government Procurement

MILITARY

MIL-STD-147	Palletized Unit Loads
MIL-STD-167-1	Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)
MIL-STD-2073-1	DOD Materiel Procedures for Development and Application of Packaging Requirements

2.1.2 Other Government documents. The following other Government document form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF TRANSPORTATION (DOT)

Code of Federal Regulations (CFR) Title 49, Part 178	Transportation

(The Code of Federal Regulations is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. Where indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

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(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 Standard Practice for Commercial Packaging; (DOD Adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B40.1 Gages – Pressure and Vacuum, Indicating Dial Type – Elastic Element

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 17A Wet Chemical Extinguishing Systems

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

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3.2 Recovered materials. Unless otherwise specified herein, all equipment, material and articles incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practical without jeopardizing the intended use. The term ""recovered materials"" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification.

3.3 Design and construction. The fire extinguishing system shall consist of cylinders, mounting brackets, aqueous potassium carbonate (APC) solution, cylinder valves, pressure gages, pressure switches, detection and actuation equipment, vent plugs, strainers and nozzles in accordance with NFPA Standard 17A and as specified herein.

3.3.1 Extinguishing agent cylinders. Class I extinguishing agent cylinders shall be in accordance with specifications 4B, or 4BW of Department of Transportation Title 49, Code of Federal Regulations, part 178 (49CFR178), and shall have a working pressure of at least 195 lbs/in^2 gage. Nominal cylinder capacities (plus or minus 5 percent) shall be 242 cubic inches and 1525 cubic inches for types A and B, respectively. Nominal cylinder diameters shall be 5 1/8 inches and 8 1/4 inches for types A and B, respectively. Class II extinguishing agent cylinders shall be in accordance with MIL-C-16310 modified to meet the requirements of this specification.

3.3.2 Extinguishing agent cylinder valves. Extinguishing agent cylinder valves for types A and B cylinders shall be the same. The cylinder valves shall be manually operable locally by lever and shall be operable remotely by cable and pneumatic pressure. Valves shall have connections for agent outlet and a pressure gage. Operating levers shall be secured by a pull-pin which is held in place by a lead and wire seal. The valve shall be fitted with a syphon tube which shall extend to a point 1/4 to 1/2 inch from the bottom of the cylinder.

3.3.3 Pressure gages. Dial type pressure gages shall be in accordance with American National Standards Institute publication ANSI B40.1 and shall be installed on extinguishing agent cylinder and pressure release control valves to monitor cylinder and release cartridge pressures. Gage dials shall have a 1-1/2 inch minimum diameter. Pressure range shall be 0 to 350 lb/in^2 (with 25-pound subdivisions on the dial face) for extinguisher cylinder valves and 0 to 600 lb/in^2 (with 50-pound subdivisions on the dial face) for pressure release cartridge control valves. Windows for pressure gages shall have heat treated glass. Pressure release cartridge control gages shall have stem mounted bottom connections. Extinguishing cylinder pressure gages shall have a center back connection and shall incorporate a corrosion resistant (CRES) bourdon tube.

3.3.4 Mounting brackets. Clamp type mounting brackets of other than the quick release type designed for welding or bolting (at the installer's option) to a bulkhead shall be supplied by the contractor. Brackets shall provide sufficient support for cylinders, cylinder valves, gages, and pressure switches, and shall meet the requirements of this specification when subjected to the tests of 4.5. Nonmagnetic (other than aluminum) brackets shall be furnished when class II equipment is specified (see 6.2.1).

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3.3.5 Switches. Pressure switch components exposed to aqueous potassium carbonate shall be corrosion resistant steel. A pressure switch shall operate on extinguisher cylinder pressure. The switch shall actuate local audible and visible alarms when cylinder pressure drops to 120 ± 5 lb/in^2, gage. A pressure switch shall acknowledge system activation when the pressure in the distribution piping, downstream of the extinguisher cylinder, reaches 40 lb/in^2, gage. The switch shall be type IC/L6C in accordance with MIL-S-16032 and shall actuate remote audible and visible alarms, de-energize heater elements, and close the grease interceptor damper.

3.3.6 Recharge/test valve. A valve shall be provided to recharge extinguishing agent cylinders with nitrogen and for bleeding off nitrogen to test the pressure switch. The valve shall be similar or equal to Schroder special tank valve having NPT male threads. The valve shall not be an integral connection to the cylinder valve or the syphon tube.

3.3.7 Potassium carbonate solution. Potassium carbonate solution shall be the composition specified in Table I. The solution shall be furnished in 2-1/2-quart or 4-gallon quantities as specified (see 6.2.1). The potassium carbonate content shall be determined as specified in 4.5.4.

TABLE I. *Composition for the potassium carbonate solution.*

Component	Percent by Weight (± 5% of percentage for each ingredient)[1]
Potassium Carbonate (K_2CO_3) (Technical Grade)	42.2
Water (H_2O)	57.8
Total	**100.0**

[1]No other ingredient shall be included.

3.3.8 Detection and actuation equipment. Detection of fire shall be by Underwriters Laboratory (UL) approved 360 °F (182.2 °C) fusible links. Links and attachments shall not be fitted with extensions or other auxiliary means of attachment. One link will be located over each fryer and within each protected duct. All links will be connected together by a taut CRES cable which also connects to a remote manual control and to the cylinder actuating mechanism. Separation of any link or pulling of a pull-pin from the manual remote control will result in relaxation of the cable and actuation of the cylinders either by release of a direct-acting spring actuator or by release of a spring actuator which releases nitrogen from a pneumatic release cartridge to open pneumatically operated cylinder valves. The pneumatic release cartridge shall be in accordance with specification 3E of 49CFR178 and shall contain nitrogen at 300 lb/in^2, gage at 70 °F (21.2 °C) to operate cylinder valves. Corner pulleys for change in direction of CRES cable shall be enclosed pulleys of a sealed, lubricated design having brass bearings. End connections of the pulleys shall have 1/2 inch electrical metallic tubing connectors.

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3.3.9 Strainers. Strainers shall be installed for each discharge nozzle. The nozzle body shall house and retain the strainer element which shall be cylindrical in shape and fabricated of CRES wire mesh capable of retaining all solid particles which will not pass through nozzle openings.

3.3.10 Nozzles. Nozzles shall be of CRES (except 202 or 302) and shall be fitted with a seal disc over the outlet end to prevent the entrance of foreign matter into the piping system. When the system is actuated, the seal discs shall rupture without restricting the flow from the nozzle.

3.3.10.1 Each nozzle shall be capable of protecting one deep fat fryer surface up to 38 x 26 inches with a maximum side dimension of 27 inches. Inlet connections shall be 1/2-14 NPT (male).

3.3.11 Cylinder recharge apparatus. Extinguishing agent cylinders, pressure release cartridge recharge apparatus and flexible high pressure outlet hose shall be supplied for refilling from nitrogen storage cylinders, RR-C-901/1. The cartridge recharge apparatus shall be in the form of a pressure regulator with an outlet relief valve installed and set at 400 lb/in^2, gage. Flex high pressure hoses shall be equipped with adapters and air chuck. The regulator capacity (in a normal operating mode or in any failed condition) shall not exceed 50 percent of the relief valve flow rate with relief valve set at 400 lb/in^2, gage.

3.3.12 Distribution system vent. A ball-check type vent, to allow breathing of the distribution system piping, shall close upon agent cylinder discharge. The vent shall incorporate 1/2-14 NPT (male) threads.

3.3.13 Shock. Components shall meet the Type A, Grade A, Class 1 shock requirements of MIL-S-901 when subjected to the test of 4.5.2.

3.3.14 Vibration. Components shall meet the Type 1 vibration requirement of MIL-STD-167-1 when subjected to the test of 4.5.3.

3.3.15 Color. Cylinders and control boxes shall be painted with red enamel in accordance with color number 11105 of FED-STD-595.

3.3.16 Magnetic permeability. Class II system components relative magnetic permeability after fabrication of all material shall not exceed 2.0 (see 4.5.5).

3.4 Tools. No special tools shall be required for installation or maintenance of the system. Special tools are defined as those tools not listed in the Federal Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Service (DCAS)).

3.5 Information plates. Information plates conforming to types A, B or G of MIL-P-15024 and normal service in accordance with MIL-P-15024/5 shall be furnished at the contractor's option. The material shall be compatible with the material on which it is mounted. One plate shall be installed at each manual operation point. Information shall be adequate to direct proper system operation and prevent inadvertent actuation.

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3.6 Certification by Underwriters' Laboratories, Inc. or Factory Mutual. Prior to submission of items for acceptance, the contractor shall submit to the contracting officer or his authorized representative, evidence that Underwriters' Laboratories (UL), Inc. or Factory Mutual (FM) has approved the equipment for extinguishment, splash, discharge (distribution and time), and cyclic life (see 6.2.2). Evidence of approval of similar equipment by UL or FM will be considered acceptable when approved by the Naval Sea Systems Command (NAVSEA).

3.7 Workmanship. Components shall be uniform in quality and evidence a level of workmanship which conforms to high-grade commercial practice for fire protection equipment. Components shall be free from irregularities, defects or foreign matter which affect their appearance or which may affect their performance, durability, reliability, or safety.

3.8 Drawings. The contractor shall prepare drawings in accordance with the data ordering documents (see 6.2.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and test) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4).

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4.3 First article inspection. First article inspection shall be performed on one component or repair part, as applicable, and shall consist of the inspections and tests specified in Table II.

TABLE II. *First article inspection.*

Test	Requirement paragraph	Test paragraph
Visual examination	3.11	4.5.1
Shock	3.3.13	4.5.2
Vibration	3.3.14	4.5.3
Potassium carbonate content	3.3.7	4.5.4

4.4 Quality conformance inspection.

4.4.1 Inspection of product for delivery. Inspection of product for delivery shall consist of the visual and dimensional examination (see 4.5.1), the potassium carbonate content test (see 4.5.4), and the inspection of packaging (see 4.4.3).

4.4.2 Sampling. When required by contractor purchase order, sampling shall consist of the sampling specified (see 6.6).

4.4.3 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing, and marking for shipment and storage shall be in accordance with the requirements of 6.6.4.

4.5 Methods of Inspection.

4.5.1 Visual and dimensional examination. When specified each sample item selected shall be visually and dimensionally examined (see 6.6.2).

4.5.2 Shock test. The shock test shall be conducted as follows: Assemblies consisting of one charged pneumatic actuator and one charged and pressurized cylinder with cylinder valve, gage and pressure switches shall be affixed to the bulkhead type mounting bracket and connected to the pressure release control by approximately 30 feet of 1/4-inch outside diameter copper tube. The tube shall be coiled and adequately supported so as not to place undue strain on the equipment. This assembly shall be subjected to a high impact shock test of MIL-S-901 for lightweight items to determine conformance with 3.3.13. Any change in parts or fasteners subsequent to the test shall require approval by NAVSEA, or retesting.

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4.5.3 Vibration. After the test of 4.5.2, the equipment shall be tested to determine conformance with 3.3.14, except that it shall not be actuated to perform its function during the test. Upon completion of the vibration test, the pressure switches shall perform as specified in 3.3.5 (except that pressure tolerance shall be plus or minus 10 lb/in^2 gage) and the pressure gages shall be accurate within 4 percent of the central 1/2 of the scale and 6 percent of the remaining 1/2 of the scale range.

4.5.4 Potassium carbonate content. The first sample selected from a lot in accordance with 6.6.3 or a first article inspection sample, shall be tested by the titration method to determine that the potassium carbonate composition is within the limits of 3.3.7, Table I. Thereafter, samples from the same lot which require testing may be tested by the specific gravity method or the titration method at the contractor's option.

4.5.4.1 Titration method. The potassium carbonate content shall be determined as follows: Weigh approximately 0.5 gram of the solution and record the weight (plus or minus 1 milligram) as S. Add 50 milliliters (mL) of distilled water and several drops of phenolphthalein indicator solution. Titrate the solution with accurately standardized 1/10 normal sulfuric acid solution to the phenolphthalein endpoint and record the volume of acid used as P. Add methyl orange indicator to the solution. Titrate with the acid to a salmon pink color and record the total volume used as M. Calculate the potassium carbonate content as follows:

$$\% \ K_2CO_3 = \frac{6.9 \ N \times 2P}{S} \text{ and } \% \ K_2C0_3 = \frac{6.9N \times M}{S}$$

Where: N = Normality of sulfuric acid
P = Volume (ml) of sulfuric acid to phenolphthalein endpoint
M = Volume (ml) of sulfuric acid to methyl orange endpoint
S = Sample weight in grams.

Both values calculated above shall be within the limits specified.

4.5.4.2 Specific gravity method. Measure the specific gravity and temperature of the sample solution with suitable, calibrated instruments. The specific gravity shall be 1.440 to 1.450 when measured at or corrected to a temperature of 60 °F (15.6 °C). This corresponds to a range of 42.3 to 43.2 percent potassium carbonate.

4.5.5 Magnetic permeability (For Class II only). The magnetic permeability of each item or assembly shall be non-magnetic in accordance with DOD-STD-2143, determined by means of a permeability indicator of the go-no go type conforming to or equal to that specified in MIL-I-17214 and shall be tested and documented in accordance with test method PO1 of DOD-STD-2142. The magnetic permeability shall conform to 3.3.16.

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5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisitions. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4.)

5.1 General.

5.1.1 Navy fire-retardant requirements.

a. *Treated lumber and plywood.* When specified (see 6.2.1), all lumber and plywood including laminated veneer material used in shipping container and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Levels A and B	–	Type II – Weather resistant Category 1 – General use
Level C	–	Type I – Non-weather resistant Category 1 – General use.

b. *Fiberboard.* Unless otherwise specified (see 6.2.1), fiberboard used in the construction of class-domestic, non-weather resistant fiberboard and cleated fiberboard boxes including interior packing forms shall meet the flamespread index and the specific optic density requirements of PPP-F-320 and amendment thereto.

5.2 Preservation. Preservation shall be Level A or commercial as specified (see 6.2.1).

5.2.1 Level A. The cleaning, drying and methods of preservation (unit protection) shall be in accordance with MIL-P-116 and as specified herein. Preservatives shall not be used.

5.2.1.1 Potassium carbonate solution. Potassium carbonate solution shall be furnished (see 3.3.7) in 2-1/2 quart and 4-gallon quantities (see 6.2.1) in containers as specified herein.

5.2.1.1.1 2-1/2-Quart container. The 2-1/2-quart quantity shall be furnished in Type II, Class B, Grade GP containers conforming to PPP-C-569 or in Type V, Class 8 cans conforming to PPP-C-96 with container selection at the option of the contractor. Cans shall have plan B coating with side seam stripping.

5.2.1.1.2 4-Gallon container. The 4-gallon quantity shall be furnished in any one of the containers listed herein with selection of the container at the option of the contractor:

a. Plastic, molded, overpacked – Type II, Grade GP, Class D of PP-C-569

b. Metal pails – Type I, Class 3, 4, 5, or 8 of PPP-P-704

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c. One piece molded plastic container – Specification No. 34 of Department of Transportation Title 49, part 178.19, Code of Federal Regulations and the following:

(1) Shall be stackable and self-supporting.

(2) Shall be provided with a threaded-type plastic cap fitted with a gasket for the pour opening.

(3) May be provided with a vent opening having an easily punctured membrane. When furnished, vent opening shall be provided with a threaded type plastic cap.

(4) Shall be provided with an integrally molded or recessible plastic or metal handle.

(5) The torque required to remove the pour opening cap shall not exceed 50-inch pounds.

Metal pails shall be provided with push-pull metal or plastic spouts. Wire handles and bails shall be zinc-coated or otherwise protectively coated to resist corrosion.

5.2.1.2 Cylinders (see 3.3.1). Cylinders shall be unit protected, method III of MIL-P-116.

5.2.1.3 System components. System components shall be preserved in accordance with MIL-E-17555. All openings shall be sealed with plastic plugs or caps to prevent entrance of foreign materials. Solution shall be packaged as specified in 5.2.1.1.

5.2.2 Commercial. Commercial packaging and unit packages shall able in accordance with ASTM D 3951.

5.3 Packing. Packing shall be level A, B, C, or commercial as specified (see 6.2.1).

5.3.1 General.

5.3.1.1 Cushioning. Unit packs of potassium carbonate solution, when over-packed for shipment, shall be provided with absorbent cushioning material which shall completely absorb the entire liquid contents resulting from container damage, rupture, or leakage. Selection of cushioning material shall be in accordance with the requirements of MIL-E-17555.

5.3.2 System parts.

5.3.2.1 Levels A, B, and C containers. System parts of one complete extinguishing system shall be packed in exterior shipping containers for the level of packing specified (see 5.3), in accordance with table VII, exterior shipping container requirements, of MIL-STD-2073-1, Appendix C, and herein. Unless otherwise specified (see 6.2.1), container selection and options shall be at the contractor's option.

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5.3.2.1.1 Caseliners, closure, and gross weight.

5.3.2.1.1.1 Caseliners. Unless otherwise specified (see 6.2.1), level A shipping containers containing extinguishers preserved commercially (see 5.2.2) shall be provided with waterproof caseliners in accordance with MIL-STD-2073-1.

5.3.2.1.1.2 Closure. Container closure, reinforcing, or banding shall be in accordance with the applicable container specification or appendix thereto except that weather-resistant fiberboard boxes shall be closed in accordance with method V and reinforced with non-metallic or tape banding and domestic or fire retardant fiberboard boxes shall be closed in accordance with method I using pressure sensitive tape.

5.3.2.1.1.3 Weight. Wood, plywood, and cleated type containers exceeding 200 pounds gross weight shall be modified by the addition of skids in accordance with MIL-STD-2073-1 and the applicable container specification or appendix thereto.

5.3.3 Solution. Unless otherwise specified (see 6.2.1), solution packaged as specified in 5.2 shall be packed for shipment in accordance with the appendix to the applicable container specification for the level of packing specified (see 5.3), except the 4-gallon containers shall not require overpacking. When specified (see 6.2.1), 4-gallon containers shall be palletized in accordance with MIL-STD-147.

5.3.4 Cylinders. Cylinders shall be packed for shipment in accordance with the requirements of RR-C-910 for the level specified (see 6.2.1).

5.3.5 Commercial. Commercial packing shall be in accordance with ASTM D 3951 and herein.

5.3.5.1 Container modification. Shipping containers exceeding 200 pounds gross weight shall be provided with a minimum of two, 3- by 4-inch nominal wood skids laid flat, or a skid- or sill-type base which will support the material and facilitate handling by mechanical handling equipment during shipment, stowage, and storage.

5.4 Marking, Levels A, B, C, and commercial. In addition to any special marking required (see 6.2.1) and herein, interior (unit) packs, shipping containers and pallet loads shall be marked including bar coding for shipment, stowage, and storage in accordance with MIL-STD-2073-1, Appendix F and the applicable container specification and appendix thereto.

5.4.1 Additional Marking (For Class II Only). Unless otherwise specified (see 6.2.1) the following special marking on the shipping container is required for class II components.

NON-MAGNETIC MATERIAL–PROTECT FROM MAGNETIC CONTAMINATION

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5.4.2 Special Marking. Underfilled standard size containers, (that is, 1- and 5-gallon containers) shall be marked indicating actual contents as 2-1/2 quarts or 4-gallons respectively. Plastic containers (see 5.2.1.1.2(c)), shall be provided with a warning stating:

DO NOT STACK MORE THAN TWO CONTAINERS HIGH

6. NOTES.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Aqueous potassium carbonate extinguishing systems are intended for protection of deep fat fryers and doughnut fryers and are used in conjunction with kitchen range equipment classified under the heading UL listed grease extractors.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification

b. Class, type, and quantity of each type cylinder installation (see 1.2)

c. Number and size of containers required (see 3.3.1 and 5.2.1.1)

d. If nonmagnetic brackets are required (see 3.3.4)

e. Number of each type of corner pulleys required (see 3.3.8)

f. Number of nozzles required (see 3.3.10)

g. When fire retardant materials are/are not required (see 5.1.1)

h. Levels of preservation, packaging, packing, and marking, other than as specified (see 5.2, 5.3, and 5.4)

i. When caseliners are not required (see 5.3.2.1.1.1)

j. When solution is to be packed other than specified (see 5.3.3)

k. When 4-gallon containers require palletization (see 5.3.3)

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l. Level of packing required for cylinders (see 5.3.4)

m. Special marking required (see 5.4).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below should be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph	DID No.	Data requirement title	Option
3.6	DI-MISC-80678	Certification Data/Report	
3.8	DI-DRPR-80651	Drawings, Engineering	
4.1.2	DI-T-2072	Test Reports	

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DOD 5000.19L, Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 Material safety data sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent government mailing addresses for submission of data are listed in Appendix B of FED-STD-313.

6.2.2.2 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of the specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article. When a first article inspection is required, the item should be a first production item, or it may be a standard production item from the contractor's current inventory as specified in 4.3. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested

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by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Sub-contracted material and parts. The packaging or preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Technical manual acquisition. This specification must be listed on the Contract Data Requirements List (DD Form 1423) in order to acquire the technical manuals described by this specification, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

6.6 Sampling plan. When a sampling plan is required the contracting officer should provide specific guidance to offerors for the number of items to be tested and the acceptance criteria. The following is a suggested sampling plan. Inspection systems should be in accordance with MIL-I-45208.

6.6.1 Inspection lot. A lot shall consist of all components of the same size and all parts which are produced on the same facilities, using identical materials, manufacturing and assembly procedures. For the potassium carbonate solution, a lot shall consist of all containers filled from the same uniform batch, regardless of size. Any addition of materials or combination of partial batches shall constitute a new batch for lot identification purposes.

6.6.2 Visual and dimensional examination. Each sample item selected should be visually and dimensionally examined to verify conformance with the requirements of this specification. Any sample containing one or more of the following defects should be rejected:

a. Type not as specified
b. Missing or incorrect parts
c. Parts loose or damaged so as to interfere with proper operation
d. Threads not as specified
e. Part identification not as specified
f. Dimensions incorrect affecting proper operation
g. Weights not as specified
h. Magnetic permeability not as specified
i. Parts loose or damaged but not interfering with operation
j. Dimensions incorrect but not interfering with operation
k. Paint or identification markings missing, incorrect, or illegible.

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6.6.2.1 **Acceptance/rejection** [1, 2, 3].

Lot or batch size	Sample size	Accept	Reject
2 to 15	*	0	1
16 to 25	*	0	1
26 to 50	20	0	1
51 to 90	20	0	1
91 to 150	20	0	1

*Indicates entire lot must be inspected

[1]All defective items must be replaced with acceptable items prior to lot acceptance.
[2]Inspect sample size until reject criteria is reached.
[3]Rejected lots may be screened and resubmitted for inspection and test.

6.6.3 Sampling for potassium carbonate content test. A random sample should be selected from each lot for the test specified in 4.5.4.

6.6.4 Sampling for inspection of packaging. For purposes of inspection, a completed package should be considered a unit of product for the packaging examination and a completed pack prepared for shipment should be considered a unit of product for the packing examination.

6.6.5 Rejected lots. If an inspection lot is rejected, the manufacturer may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots should be inspected using tightened inspection and should not thereafter be tendered for acceptance unless the former rejection or requirement of correction is disclosed. Such lots should be separate from new lots and should be clearly identified as reinspected lots.

6.7 Subject term (key word) listing.

Deep fat fryer
Doughnut fryer
Fire protection
Galley equipment
Saponification

6.8 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing Activity
Navy – SH
(Project 4210–N233)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-E-24416B	8 February 1991

3. DOCUMENT TITLE

EXTINGUISHING SYSTEM, FIRE, FIXED, ACQUEOUS POTASSIUM CARBONATE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Technical point of contact, (Contaact Mr. John Forney (SEA 5/422) PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	(1) Commercial TPCO: 703-602-0143	(2) AUTOVON 8-332-0143
c. ADDRESS (Include Zip Code) Commander, Naval Sea Systems Command Department of the Navy(SEA 55Z3) Washington, DC 20362-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/290

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